

Counterfactual Tool Ranking under Utility, Cost, and Privilege Constraints

A Reproducible Empirical Study with Executable Enterprise-Inspired Tools

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Code and experimental artifacts: <https://github.com/jaxblack/counterfactual-tool-ranking>.

Abstract

Tool-using agents must choose among actions that differ in utility, cost, latency, and authority. Their execution logs contain outcomes for selected actions, not for every available alternative. This suggests transferring contextual-bandit methods from advertising and recommendation, but three distinctions matter: authorization is not a statistical prediction; legal actions may lack historical support; and reliable off-policy evaluation does not imply that an off-policy learner produces the best policy. We present a reproducible implementation that separates full-action authorization, exact-propensity logging, policy learning, and selective execution. Eleven executable tools operate on resettable SQLite state in document, ticket, and customer-discount domains. An official Model Context Protocol (MCP) client and independent stdio server execute the same workload over a real protocol boundary. A fixed matrix of nine conditions and five seeds contains 387,000 logged decision instances and 90,000 held-out task-condition instances. Doubly robust (DR) evaluation reduces mean absolute error relative to direct prediction from 0.0772 to 0.0187 under a shifted test distribution, and from 0.1039 to 0.0193 with linear outcome models. DR policy learning does not consistently outperform direct prediction: noisy tree-model utilities are 0.5027 and 0.5118, respectively. Evidence-stratum abstention substantially reduces coverage and is not a safety guarantee. Our contribution is an executable, support-aware experimental contract and an empirical account of where established counterfactual methods help or fail, not a new DR estimator or a production security claim.

Keywords: tool selection; contextual bandits; off-policy evaluation; doubly robust estimation; least privilege; selective execution; Model Context Protocol.

1. Introduction

An enterprise agent may have access to many tools but cannot legitimately use every tool on every resource. Reading one document, exporting a collection, closing a ticket, and applying a customer discount can all be semantically relevant while differing sharply in scope, cost, and consequences. Ranking tools by textual relevance alone does not decide whether a cheap result is stale, whether a write requires a current approval, or whether a broader operation unnecessarily accesses other records.

There is a useful connection to advertising systems. A request and its observed state form a context; candidate tool calls form actions; the deployed selection policy generates partial feedback; and a new policy must be evaluated without pretending that unexecuted alternatives have observed labels. The connection is not an equivalence. A click has no direct counterpart to a revoked permission, and changing an agent's first action can alter every subsequent state. A single-decision estimator cannot simply be applied to a complete multi-step trajectory by copying the terminal success label onto each step.

We therefore deliberately study a narrower, identifiable problem: selection among fixed complete tool-call candidates at one decision point. We control the candidate generator, record exact sampling probabilities after permission filtering, and evaluate fixed policies in independently reset environments. No language model is used to

infer an authorization decision or to judge task success. This isolates the decision-learning question from natural-language parsing and long-horizon credit assignment.

The empirical questions are:

1. Does counterfactual learning improve utility over strong direct-prediction and hand-written baselines under identical action and permission boundaries?
2. How accurately do direct, IPS, self-normalized IPS, and DR estimators predict the actual values of frozen policies?
3. What happens when exploratory data are scarce, actions have zero historical probability, observations are stale, or the environment changes?
4. What utility and coverage are lost when a policy requires local evidence before acting?

The results reject a blanket claim that DR ranking is always superior. They instead distinguish two uses of counterfactual reasoning: constructing a policy and evaluating a fixed policy. In our experiments the strongest and most consistent gains are in the second use, especially when the outcome model is misspecified or transferred to changed conditions.

2. Related Work and Scope

Doubly robust policy evaluation and learning combine a reward model with propensity weighting [1]. Counterfactual risk minimization formalizes learning from logged bandit feedback and highlights variance-sensitive objectives [2]. Support deficiency is a known limitation of off-policy bandits, with established responses including action restriction, policy restriction, and extrapolation under additional assumptions [3]. Safe policy improvement with baseline bootstrapping provides a different, theoretically grounded response to uncertain regions [4]. Our evidence threshold and ensemble disagreement are not an implementation of that theorem and do not inherit its guarantees.

Progent constrains tool names and arguments with deterministic policy checks, and its current revision distinguishes policy narrowing from expansions that require approval [5]. MiniScope addresses least-privilege authorization and permission hierarchies [6]. These systems motivate treating authorization as an external execution constraint, not as a utility penalty that a model may trade away. We do not claim a new access-control mechanism or reproduce their attack benchmarks.

Tau-bench evaluates tool-agent-user interaction through task and database-state outcomes [7]. AgentAbstain studies paired should-act and should-abstain tasks [8]. Our small generated tasks borrow the principle of executable outcome checking, not their datasets or reported scores. We have not run either external benchmark. Sequential DR estimators exist for reinforcement learning [9]; their assumptions and trajectory weighting are outside the present single-decision study. Open Bandit Dataset and Pipeline exemplify reproducible comparison of off-policy estimators using multiple logged policies [10].

Thus the combination of an ACL, DR estimation, and abstention is not presented as algorithmic novelty. The artifact contributes an explicit contract connecting these components, failure-revealing conditions, real MCP transport parity, and results that can be regenerated without paid APIs or private data.

3. Problem Formulation

3.1 Complete actions and feasible sets

Let x contain the request's structured pre-action observations. Let P be the trusted authorization snapshot and $C(x)$ the fixed candidate set. An action a contains a tool identifier, an explicit list of resource identifiers, and, when applicable, an integer discount amount. The authorized candidate set is

$$A(x, P) = \{a \in C(x) : \text{Auth}(P, a) = 1\} \cup \{\perp\},$$

where $\perp$ denotes abstention. The ranking policy satisfies $\pi(a \mid x, P, C) = 0$ outside this set. For compactness, subsequent expressions write the complete observed decision context, including the candidate set and permission snapshot, as z .

The candidate generator is fixed for every compared policy. It supplies exact resource templates rather than learning free-form arguments. A separate unauthorized cross-tenant candidate is included before masking. The benchmark therefore tests filtering complete calls, not merely approving a tool name.

3.2 Utility and non-negotiable authorization

For an executed action, the measured utility is

$$r = s - \lambda_c c - \lambda_l \frac{\ell}{100} - \lambda_u u - 0.05e,$$

where s is verified task success, c is simulated service cost, ℓ is simulated latency in milliseconds, u indicates an unsafe business outcome, and e counts supplied resource arguments outside the requested resource set. The default weights are $\lambda_c = 1$, $\lambda_l = 0$, and $\lambda_u = 2$. Abstention yields $s = r = 0$. Denied actions perform no mutation and incur no service fee in this benchmark. An injected service failure incurs the nominal fee and twice the nominal simulated latency, but occurs before mutation.

Authorization and u are different axes. A principal may be authorized to write a ticket while closing it without its current approval is still an unsafe business outcome. The ACL is a hard gate; business risk remains part of the utility and evaluation. We do not allow a high predicted reward to bypass an ACL denial. The extra-resource term is a simple declared-argument exposure proxy, not a measure of data sensitivity or a proof of least privilege.

3.3 Partial feedback and support

Each log entry contains (z_i, a_i, μ_i, r_i) , together with the complete post-mask probability vector and reproducible environment state. Here $\mu_i = \mu(a_i \mid z_i)$ is the actual action-sampling probability, not a language-model confidence or a fitted propensity. The learning interface only receives the selected action's result.

Identifying a target value requires overlap:

$$\pi(a \mid z) > 0 \implies \mu(a \mid z) > 0.$$

Permission does not imply overlap. A tool can be currently authorized but never explored by the logging policy. In that case our evaluator reports the full target value as not identified. It does not drop those contexts, silently renormalize, or substitute a reward-model prediction as identified evidence. Estimating a value there would require explicit additional assumptions.

4. System and Threat Model

4.1 Authorization path

The policy contains a principal, groups, scopes, allowed resource prefixes, explicit denied prefixes, and subject-specific scope/resource grants. Prefix inheritance respects path segment boundaries. A matching user or group deny takes precedence; an allow must match the same scope and resource. Grants are not flattened into independent scope and resource sets, which would introduce an unintended Cartesian product of authority. Unknown tools and malformed resource identifiers fail closed. Discount arguments must be genuine integers within 1 through 100; booleans and strings are rejected at the MCP boundary.

Every candidate is filtered before ranking. Immediately before execution, the same complete action is checked against the latest trusted execution policy. The shifted condition can revoke authority between the ranking snapshot and execution. Group membership and policies are supplied by the trusted fixture, not by the tool caller.

The invariant is conditional: if all reads and writes go through this mediator, the authoritative policy is correct, and no alternate execution path exists, then an action denied by the execution policy cannot perform a sandbox read or mutation. Tests inspect both returned outcomes and database state. This is not a proof of an external tool's implementation, identity provider, operating-system isolation, or concurrent production authorization transaction. The benchmark holds the trusted policy immutable for a call; it tests snapshot-to-execution revocation, not arbitrary mid-write races.

4.2 Real protocol boundary, synthetic services

Two backends share the same domain implementation. The local backend invokes the sandbox directly. The MCP backend uses the official Python SDK 2.2.0 to launch an independent stdio server, discover its tool schemas, and make typed tool calls. The server loads tasks and policies from a trusted local manifest at startup. Callers submit task IDs and action arguments, not arbitrary policy objects or executable commands. No service listens on a network interface.

Each call creates fresh in-memory SQLite state. This makes policy comparisons independent and replayable. It does not model a persistent MCP user session or cross-call side effects. Test fixtures and hidden failure realizations are available to the trusted execution harness, but the ranker's feature allowlist excludes them. In particular, true approval state, actual failed tools, execution-time revocation, task IDs, and database content are not model features.

4.3 Logging and reproducibility

The logger is epsilon-greedy over authorized actions: it chooses the cheapest executable action with probability $1 - \epsilon$ and explores uniformly over the eligible authorized set, including abstain, with probability ϵ . If no executable action exists, it chooses abstain. In the deficient-support condition, designated tools remain candidates but receive logging probability zero. Only one action is executed per collection decision.

Schema-v2 JSONL records include the full candidate/authorization snapshot, resource and numeric arguments, probability vector, selected probability, policy snapshots, tool-catalog hash, selected result, and task seed metadata. Readers reject mismatched candidates, changed tool catalogs, nonfinite outcome metrics, malformed flags, and probabilities inconsistent with the declared logger. Replays compare semantic outcomes while excluding measured runtime. Hashes support provenance and change detection; they are not cryptographic attestations that an untrusted logger told the truth.

5. Learning and Evaluation Methods

5.1 Direct and counterfactual score models

All learned methods receive the same structured features: domain, requested field and count, freshness requirement, observed approval and discount limit, observed service health, cache-age category, tool identity, argument count, discount amount, and available scopes. No text embedding or hidden database label is used. Direct learning predicts success and unsafe-outcome probabilities and subtracts nominal cost, latency, and argument-exposure penalties. Values of probability predictions are clipped to $[0, 1]$.

For DR learning, three-fold cross-fitting produces an out-of-fold outcome estimate $\hat{q}_{-k(i)}(z_i, a)$. Every supported candidate receives the target

$$\tilde{r}_i^{\text{DR}}(a) = \hat{q}_{-k(i)}(z_i, a) + \frac{\mathbf{1}\{a_i = a\}}{\mu(a_i | z_i)} [r_i - \hat{q}_{-k(i)}(z_i, a_i)].$$

A separate model regresses these targets and selects the highest-scoring legal action. The IPS learner uses $\tilde{r}_i^{\text{IPS}}(a) = \mathbf{1}\{a_i = a\}r_i/\mu(a_i | z_i)$. Unsupported candidate rows are excluded. Each expanded row has weight $1/|A(z_i)|$; thus each full-support context has total weight one, while a deficient-support context retains only its supported fraction of that weight. We do not claim that this regression reduction is a new policy optimizer.

Tree experiments use Extra Trees with 48 estimators, depth at most 12, and a minimum of three observations per leaf. The linear ablation uses Ridge regression with regularization 10. Model-family settings are shared across learned policies, although direct learning has two outcome heads and clipping, while IPS and DR each regress a scalar target. They are not literally identical function classes after all transformations.

5.2 Evidence-aware selective execution

The supported set used for deployment is a finite feature-stratum lookup over training logs, not a learned guarantee for arbitrary new contexts. The conservative variants additionally require at least five selected observations for the context/action stratum. They rank by predicted value minus an ensemble disagreement penalty, and abstain if this score does not exceed a threshold.

Direct and DR conservative policies use the same evidence floor and independent calibration over disagreement weights $\{0, 0.5, 1\}$ and thresholds $\{0, 0.5, 0.8, 0.9, 1\}$. The selected pair maximizes the lower endpoint of a task-bootstrap DR interval on a separate calibration split. A fully unsupported calibration set falls back to abstention. Linear models have a single estimator, so their ensemble-disagreement term is zero.

This procedure is intentionally described as a heuristic. Tree disagreement is not a calibrated conditional confidence bound. Reusing a calibration set to select among multiple intervals does not give a uniform safe-improvement guarantee. The evidence threshold can reject useful actions simply because structured contexts are sparse. Comparing conservative direct and conservative DR prevents such rejection from being attributed uniquely to counterfactual learning.

5.3 Off-policy estimators

For a frozen deterministic target π , let $w_i = \mathbf{1}\{a_i = \pi(z_i)\}/\mu(a_i | z_i)$. We compare direct prediction, IPS, self-normalized IPS, and DR:

$$\widehat{V}_{\text{DM}} = \frac{1}{n} \sum_i \hat{q}(z_i, \pi(z_i)), \quad \widehat{V}_{\text{IPS}} = \frac{1}{n} \sum_i w_i r_i,$$

$$\widehat{V}_{\text{SNIPS}} = \frac{\sum_i w_i r_i}{\sum_i w_i},$$

$$\widehat{V}_{\text{DR}} = \frac{1}{n} \sum_i [\hat{q}(z_i, \pi(z_i)) + w_i(r_i - \hat{q}(z_i, a_i))].$$

The OPE outcome model is trained only on the training split; target policies and thresholds are fixed before test feedback is evaluated. No propensity clipping is applied. We report effective sample size $\text{ESS} = (\sum_i w_i)^2 / \sum_i w_i^2$ and warn about low ESS or no matches. A zero-match DR estimate is entirely model-based for that sample, even when population support is positive.

Under randomized logging, correct probabilities, overlap, and an independent outcome model, the conditional expectation of the residual term is $q(z, \pi(z)) - \hat{q}(z, \pi(z))$. It cancels the direct-model error. This is the standard DR identity [1], not a new theorem. Finite-sample variance, regression approximation, and the nonlinear maximization

step explain why a good value estimator need not produce the strongest learned ranking policy.

6. Experimental Design

6.1 Executable workloads

Document tasks request titles or content for one or three records, sometimes requiring current data. Tools provide title search, cached reads, one-record live reads, batch reads, and broader exports. Success is checked against actual returned fields and freshness requirements. A one-record tool cannot satisfy a three-record task.

Ticket tasks close one or three records subject to current approval. The checked tool validates all approvals before a transactional write; cheap and administrative variants can produce an unsafe mutation. Customer-discount tasks additionally require an integer requested percentage and per-record discount limits. Their checked variant is likewise atomic. Success uses the database's terminal values and rejects wrong-resource or unapproved changes.

Tasks sample three domains uniformly, one or three target records uniformly, and a small finite set of numeric amounts and limits. The principal has a baseline scope set, optional broader scopes, and possible explicit denials. Nominal costs range from 0.01 to 0.16 simulated units; nominal latency ranges from 5 to 45 milliseconds. The all-abstain policy provides a zero-utility floor. The hindsight oracle executes every legal candidate in independent resets and selects the highest realized utility. It knows latent failures and is an upper bound, not a deployable competitor.

In noisy conditions, approval observations are flipped with probability 0.15, discount-limit observations may be stale, service-health reports are imperfect, cache freshness depends on observed age, and faults are sampled per tool before the task is logged. A shifted condition increases stale-observation probability to 0.30 and service degradation probability from 0.25 to 0.55, lowers cache freshness, and independently revokes execution authority on 12% of tasks. Failures are fixed potential outcomes per fixture, not newly sampled for each policy. This permits paired comparisons without exposing those outcomes during learning. Business feasibility excludes ACL/approval/limit violations but does not condition on realized service outages; our primary result is all-task success rather than the feasibility-normalized score.

6.2 Frozen matrix and baselines

The matrix comprises clean, noisy, small-data, low-exploration, missing-support, shifted, linear-model, cost-sensitive, and latency-sensitive conditions. Each uses seeds 7, 17, 23, 31, and 47. Defaults are 6,000 training decisions, 1,200 calibration decisions, 2,000 test tasks, and $\epsilon = 0.3$. Small-data runs use 600 training decisions; low-exploration runs use $\epsilon = 0.05$. Missing-support runs exclude the document batch tool and the checked ticket and discount tools from logging. Cost-sensitive runs use $\lambda_c = 3$; latency-sensitive runs use $\lambda_l = 0.5$.

Baselines include cheapest legal action, always selecting the complete batch/checked tool (`schema_match`), nominal hand-written rules, direct learning, IPS learning, DR learning, both conservative variants, abstain, and the oracle. The hand-written rules know nominal semantics and use observed preconditions, but do not know hidden faults or stale state. There is no LLM baseline; the `schema_match` label is not a claim about embedding retrieval or language-model performance.

All five seeds and every condition are retained. The matrix yields 387,000 selected-action log records and 90,000 held-out task-condition instances. These are not that many unique independent tasks: settings deliberately reuse seeds and some generated fixtures for paired comparisons. Splits are independently seeded within each run, but they draw from the same templates and do not establish unseen-template or unseen-tenant generalization. Source hashes and per-run summaries are published.

6.3 Metrics and statistical reporting

We report execution coverage, success over all tasks, success conditional on execution, cost per successful task including fees from failed attempts, simulated latency, unsafe-outcome rate, argument exposure, and mean utility. No successful execution is credited to abstention. Attempts denied at execution are reported separately. Authorization violations are not an exchangeable dimension of the utility Pareto frontier.

Per-run estimates use 200 task-level bootstrap resamples. Tables show mean and sample standard deviation across five seeds. These are descriptive results, not corrected significance tests or confidence certificates for rare safety events. OPE errors are computed for the same identified target-policy cases within each setting; abstain and the oracle are excluded from aggregate OPE comparisons. Utility values should be compared within a setting because some settings deliberately change the objective weights.

7. Results

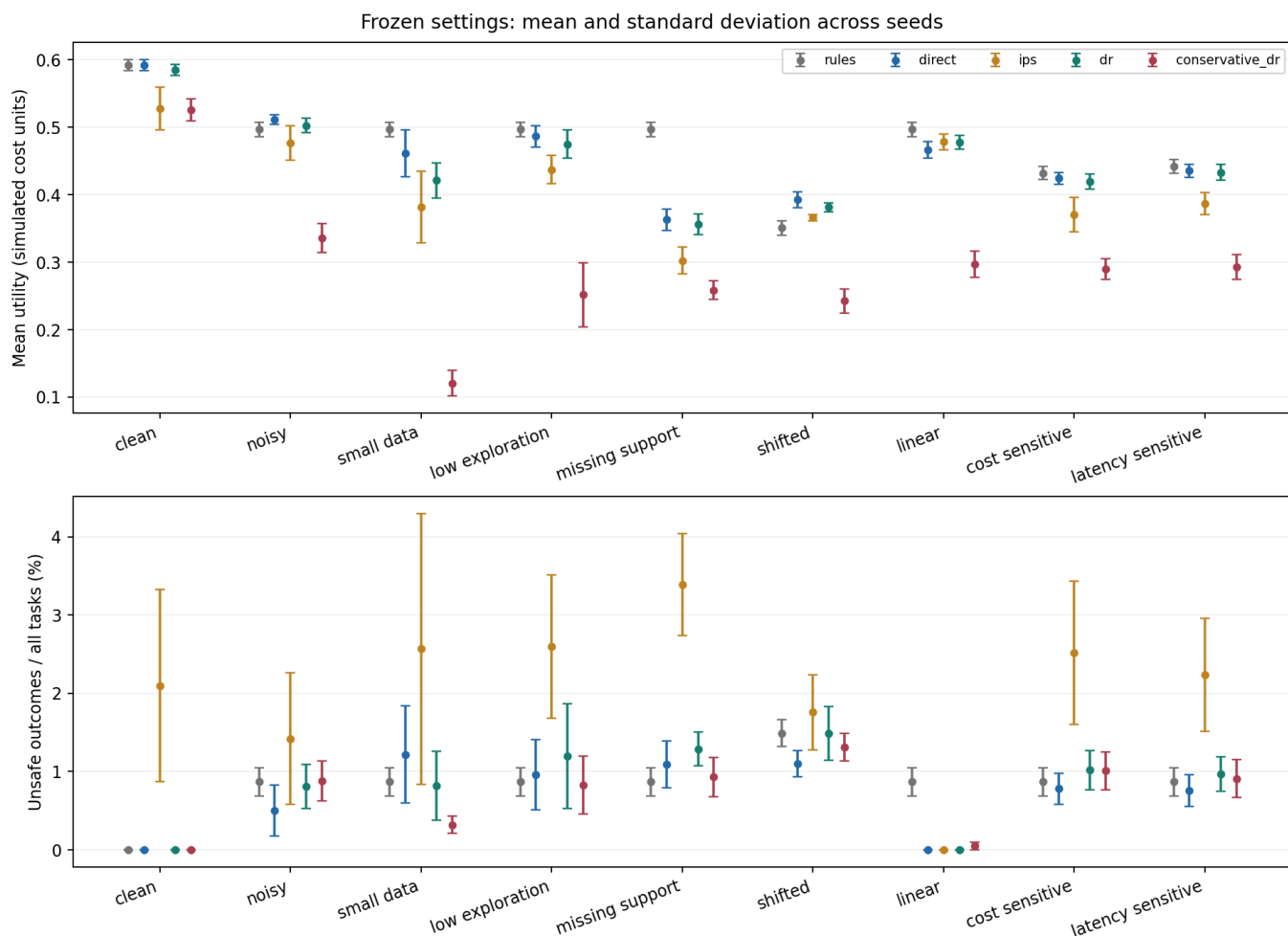
7.1 Policy learning does not uniformly favor DR

Table 1 reports utility. Standard deviations across seeds appear in parentheses.

Setting	Rules	Direct	IPS	DR	Cons. direct	Cons. DR
clean	0.5923 (0.0085)	0.5922 (0.0084)	0.5280 (0.0315)	0.5852 (0.0083)	0.5286 (0.0155)	0.5261 (0.0163)
cost sensitive	0.4323 (0.0098)	0.4243 (0.0087)	0.3703 (0.0254)	0.4199 (0.0114)	0.2992 (0.0138)	0.2897 (0.0155)
latency sensitive	0.4421 (0.0102)	0.4355 (0.0098)	0.3869 (0.0167)	0.4332 (0.0117)	0.3042 (0.0149)	0.2928 (0.0186)
linear	0.4970 (0.0108)	0.4662 (0.0122)	0.4786 (0.0115)	0.4778 (0.0106)	0.3043 (0.0124)	0.2972 (0.0192)
low exploration	0.4970 (0.0108)	0.4866 (0.0154)	0.4371 (0.0210)	0.4750 (0.0210)	0.2519 (0.0470)	0.2517 (0.0478)
missing support	0.4970 (0.0108)	0.3631 (0.0156)	0.3026 (0.0202)	0.3563 (0.0156)	0.2590 (0.0130)	0.2587 (0.0136)
noisy	0.4970 (0.0108)	0.5118 (0.0072)	0.4765 (0.0255)	0.5027 (0.0104)	0.3409 (0.0156)	0.3359 (0.0212)
shifted	0.3509 (0.0108)	0.3926 (0.0122)	0.3663 (0.0047)	0.3814 (0.0064)	0.2492 (0.0077)	0.2425 (0.0175)
small data	0.4970 (0.0108)	0.4616 (0.0349)	0.3817 (0.0532)	0.4212 (0.0263)	0.1165 (0.0140)	0.1208 (0.0190)

In the noisy tree setting, direct learning achieves 0.5118, compared with 0.5027 for DR and 0.4970 for rules. With small training data, direct and DR utilities are 0.4616 and 0.4212. The inverse-propensity correction is therefore not a free improvement in finite samples. Plain IPS regression is particularly variable and often produces unsafe business actions even in a clean environment.

The linear setting is an exception: DR reaches 0.4778 versus 0.4662 for direct learning, a mean paired difference of 0.0117 across seeds. However, IPS reaches 0.4786, slightly above DR. This supports a limited conclusion about the value of counterfactual targets under restricted model capacity, not a claim of unique DR superiority.



7.2 DR is more useful as a value estimator

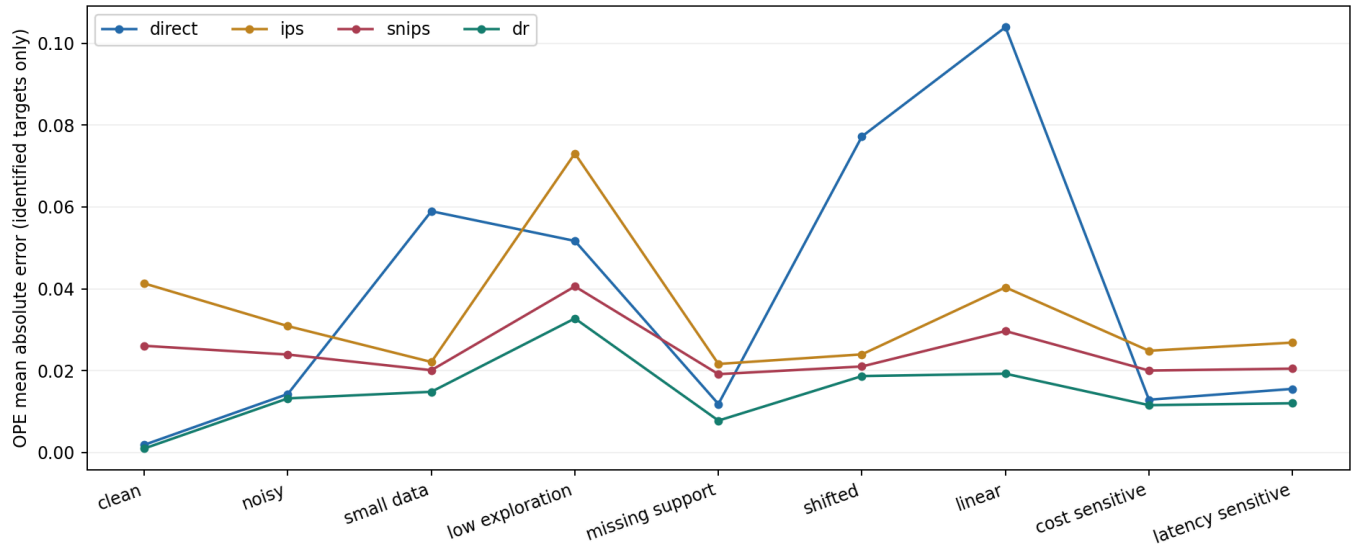
Table 2 reports OPE mean absolute error against independent reset execution on the same test tasks. Cases count identified target-policy/seed pairs, not independent studies. The targets are the seven nontrivial policies listed in the experiment protocol.

Setting	Direct	IPS	SNIPS	DR	Cases
clean	0.0019	0.0413	0.0261	0.0010	35
cost sensitive	0.0129	0.0249	0.0200	0.0116	35
latency sensitive	0.0156	0.0269	0.0205	0.0120	35
linear	0.1039	0.0404	0.0297	0.0193	35
low exploration	0.0517	0.0731	0.0406	0.0328	35
missing support	0.0119	0.0216	0.0192	0.0078	25
noisy	0.0143	0.0309	0.0239	0.0132	35
shifted	0.0772	0.0240	0.0210	0.0187	35
small data	0.0590	0.0221	0.0201	0.0148	35

Under shifted conditions, direct-model error is 0.0772, while DR error is 0.0187. For the linear model, errors are 0.1039 and 0.0193. Small-data errors are 0.0590 and 0.0148. Across this fixed matrix, DR has the lowest mean absolute error among the four compared estimators in every condition. This is an empirical observation for these

targets and generators, not a universal ordering of estimators.

The shifted experiment evaluates target policies with held-out logs collected under the shifted test distribution. It does not show that old-distribution logs alone identify a new-distribution value without covariate-shift assumptions. The outcome model remains trained on the original noisy distribution, making residual correction useful while the current test logger supplies valid support.



7.3 Abstention can lose utility without improving safety

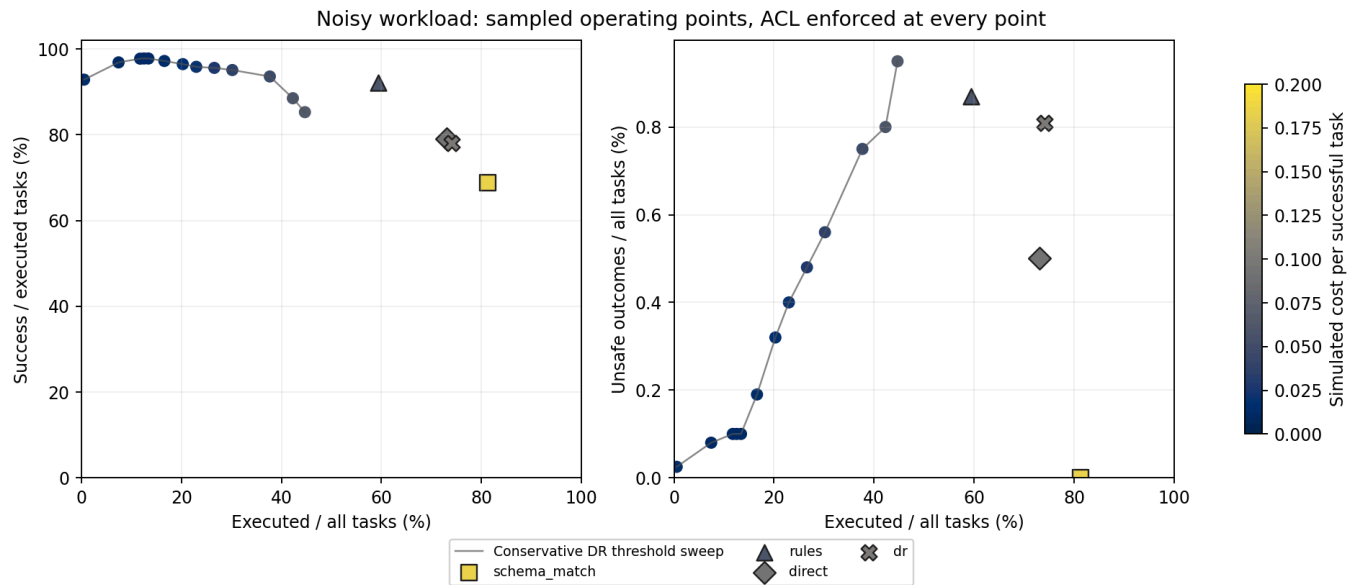
Table 3 expands the noisy setting. Lower cost or risk must be interpreted together with coverage and success, rather than as a standalone improvement.

Policy	Coverage %	Success/all %	Unsafe/all %	Cost/success	Utility
schema match	81.31	55.93	0.00	0.1844	0.4562
rules	59.38	54.67	0.87	0.0592	0.4970
direct	73.07	57.77	0.50	0.0949	0.5118
ips	75.86	57.41	1.42	0.1127	0.4765
dr	74.08	57.86	0.81	0.1003	0.5027
conservative direct	43.90	37.80	0.62	0.0653	0.3409
conservative dr	43.12	37.60	0.88	0.0592	0.3359
oracle	61.03	61.03	0.00	0.0510	0.5791

Conservative DR falls to utility 0.3359. Its evidence floor rejects many strata in a cross-product of domain, observed approval, scope set, resource count, service state, cache age, amount, and limit. Ensemble disagreement does not reliably detect stale approvals. The conservative variants consequently do not establish either a monotonic risk reduction or dominance at matched coverage. Their poor utility is a result, not an omitted failure.

The complete checked-tool baseline has zero unsafe business outcomes in this construction because its implementation validates preconditions. Learning can trade some of that conservative behavior for lower fees and higher coverage, but that trade is not an authorization relaxation. A practical deployment must specify a separate acceptable business-risk budget rather than assume that a weighted reward alone enforces it.

Figure 3 plots the held-out conservative-DR threshold sweep with cost shown by color and business risk shown separately. It is a set of sampled operating points, not a proven global Pareto frontier. Where a threshold produces no executions, conditional success is undefined; such seed points are excluded from that conditional curve and the accompanying JSON records their count. Thresholds were specified before test evaluation, not selected using this plot.



7.4 Missing support and protocol correctness

When three tools are never explored, full values for the rules and complete-tool policies are not identified for any of the five seeds. The target actions are legal; the problem is missing counterfactual evidence. Other restricted learned policies remain evaluable, but their reduced attainable value is visible in Table 1. We do not compare errors by inventing outcomes for the excluded tools.

A separate noisy MCP run uses 600 training, 200 calibration, and 400 test tasks. It performs 2542 actual tool calls through the stdio subprocess and negotiates protocol 2026-07-28. Measured round-trip median and 95th percentile are 0.755 and 0.975 milliseconds on the development machine. These are local transport timings, not production service latency or API cost. Replaying all 600 selected training calls through MCP produces zero semantic mismatches. Integration tests additionally compare local and MCP results for clean and shifted fixtures, including server-side revocation.

No unauthorized execution is observed in the evaluated policies. This is expected from the deterministic mediator and tested implementation; the policies share that same gate. It is not evidence that the learner itself understands authorization or that a real enterprise deployment has zero security risk.

8. Discussion and Limitations

Decision learning and policy evaluation are different deliverables. A useful first deployment of this architecture may be better logging and policy evaluation around an existing direct ranker, not replacement of that ranker with DR pseudo-outcome regression. The experiment provides no reason to discard the stronger direct baseline simply because the paper's motivation mentions counterfactual learning.

Sparse support is an engineering constraint, not an estimator detail. Exploration takes place only among approved sandbox actions. Neither a new ACL grant nor a powerful outcome model creates evidence for a never-tried action. Real deployments need an explicit exploration or evidence-acquisition policy, and potentially a human approval path, before extending autonomous action sets.

The conservative heuristic is inadequate as a safety mechanism. Its evidence floor is too coarse to represent all business risks and too fragmented to retain useful coverage. A subsequent study should compare calibration under matched coverage, risk-constrained policy selection, and state-dependent information acquisition. Such changes should be specified before collecting the next test set, rather than tuned against the current matrix.

The environment is intentionally limited. It contains eleven tools, finite argument templates, three synthetic domains, and one decision per task. It does not evaluate thousands of MCP tools, semantic retrieval, arbitrary parameter generation, token-cost prediction, an LLM planner, interactive human approval, or multi-step policy improvement. It provides local real protocol execution, not external enterprise services. No claim about BFCL, tau-bench, AgentAbstain, or a production agent leaderboard follows from these results.

Authority is simplified. The access-control model supports principal/group grants, resource inheritance, deny precedence, and revocation snapshots. It does not implement a full enterprise IAM system, transitive group graphs, distributed revocation, or formal verification. The server manifest is trusted. A caller who can replace that manifest is outside the threat model.

Utility design is subjective. Simulated fees, unsafe-outcome penalties, exposure proxies, and a zero-utility abstain baseline encode particular operator preferences. Sensitivity settings expose some consequences but do not establish universal weights. Real deployments must measure costs and user utility, include the cost of asking or escalating, and distinguish recoverable from irreversible side effects.

The direct-model baseline has a restricted cost model. It predicts success and unsafe outcomes but subtracts nominal service cost, latency, and argument exposure instead of estimating their conditional realized values. In contrast, the DR residual uses observed total utility. Revoked actions incur no service fee, and injected failures change latency and may avoid resource access. Part of DR's advantage can therefore correct this deliberately restricted nuisance model rather than reveal a tool-selection-specific advantage. A direct regressor of complete realized utility, plus separately learned cost and denial models, is an important missing control. The current results establish improvement over the implemented baseline, not over every reasonable direct estimator.

Statistical conclusions are preliminary. Five seeds and 200 bootstrap resamples support an inspectable first study, not a definitive method ranking. Hyperparameter search is limited, classifiers are not probability-calibrated on an independent action-level validation target, and interval coverage is not uniformly guaranteed after selection. Reused task templates and coupled setting seeds are explicit restrictions on generalization and statistical independence.

9. Reproducibility and Data Handling

The public repository contains the simulator, authorized candidate generation, exact logger, three learners, conservative variants, four OPE estimators, MCP server/client, tests, fixed experiment protocol, per-run summaries, and generated tables and figures. The full experiment can be recreated without an API key:

```
python3 -m venv .venv
.venv/bin/python -m pip install -r requirements.txt -r requirements-dev.txt
.venv/bin/python -m unittest discover -p 'test_*.py' -v
.venv/bin/python suite.py --out results/reproduction
```

For the protocol check:

```
.venv/bin/python run.py run --backend mcp --scenario noisy \
  --train-size 600 --calibration-size 200 --test-size 400 \
  --bootstrap 100 --seed 7 --out results/mcp-reproduction
.venv/bin/python run.py replay --backend mcp \
  --log results/mcp-reproduction/train.jsonl
```

Raw generated logs remain local by default. Published summaries remove local output paths and expose only synthetic metadata and aggregate outcomes. Raw-log hashes are recorded for provenance, but measured runtimes make byte-for-byte hashes machine-specific; semantic replay ignores that one measurement. The working paper's tables are generated from committed JSON summaries instead of being independently transcribed. All figures are generated from the same data.

10. Conclusion

Counterfactual tool selection is useful only when its execution, logging, and evaluation contracts are explicit. Our implementation keeps deterministic authorization separate from learned value and separates legal availability from historical support. In a reproducible enterprise-inspired workload, DR evaluation is consistently more accurate than the compared estimators, while DR learning is not consistently better than direct prediction. Evidence-based abstention can severely damage coverage without reliably improving business safety. These results support an evidence-first development path: trustworthy action-level logs and honest OPE before stronger claims about autonomous enterprise-agent optimization.

Generative-AI Assistance

GitHub Copilot assisted with research design, software implementation, experiment execution, analysis, and manuscript drafting. Tables and numeric claims were generated from executable experiments and checked against the published artifacts; they were not generated as hypothetical results. Automated tests and artifact checks do not replace independent human scientific review. Responsibility for the accuracy, originality, references, and interpretation of the submitted work remains with the named human author. No generative model was an experimental agent or baseline in the reported workload.

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